ℹ Exam info

Exam
Faculty of Science and Technology

Course code: DAT540

Course title: Introduction to data science

Year and semester: 2025 -Fall

Date: 27.11.2025

Duration: 3 hours

Legal aids: None

All exams submitted after the expired upload time will not be registered.
Questions - All Questions will be graded after the exam. Therefore, you will NOT receive any points immediately after completion.

Multiple Choice (40 questions): You get 1 point for each correct answer to the multiple choice questions. You will get 0 points for incorrect answers or unattempted questions.

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***NOTE: I am fully available from 9.00 to 10.00. After 10.00 it might be with a short delay.**

1 **Q1**

What is the output of A.size for the following NumPy array?

```
A = np.array([[1, 2, 3],[4, 5, 6]])
print(A.size)
```

Select one alternative:

- 6 ✔
- 3
- (2,3)
- 2

Maks poeng: 1

2 **Q2**

Which code snippet produces the output {'x', 'y', 'z'} (order may vary)?
Select one alternative:

- set("zzyxx") ✔
- list("xyz")
- tuple("xyz")
- None of the options

Maks poeng: 1

3 Q3

What will be the result of print statement?

```
arr = np.array([[10, 20, 30],
                [ 40, 50, 60],
                [70, 80, 90]])
print(arr[[0, 2], [1, 2]])
```

Select one alternative:

- [20, 90] ✔
- [[20, 90]]
- [[0, 2], [1, 2]]
- [10, 80]

Maks poeng: 1

4 Q4

What command can produce this result: pd.Series([2.0,5.0], index=['A','B'])
out of the following df?

df = pd.DataFrame({'A':[1,2,3], 'B':[4,5,6]})

1. df.mean()
2. df.mean(axis=1)
3. df.apply(np.mean)

Select one alternative:

- 2
- 1 & 3 ✔
- 1
- 3

Maks poeng: 1

5 Q5

You need to reorder the rows of a DataFrame df in ascending order according to their index labels.

Which command accomplishes this?

Select one alternative:

- df.sort_index() ✔
- df.sort()
- df.sort_index(axis=1)
- df.sort_values()

Maks poeng: 1

6 Q6

What will be the result of calling my_function?

```
def my_fucntion(data):
    return [name for name, score in data if score > 80]
students = [('Alice', 75), ('Bob', 90), ('Carol', 95), ('Dave', 60)]
print(my_fucntion(students))
✓ 0.0s
```

Select one alternative:

- ['Bob', 'Carol'] ✔
- ['Alice', 'Bob', 'Carol', 'Dave']
- {'Bob': 90, 'Carol': 95}
- [90, 95]

Maks poeng: 1

7 **Q7**

Given the following array, what will be printed?

```
my_array = [['Data', 'Science', 'Python'], [10, 20, 30, 40]]
print(my_array[0][2][::2])
```

Select one alternative:

- 'Sin'
- 'Pto' ✔
- 'nhy'
- 'thon'

Maks poeng: 1

8 **Q8**

What does the identity() function in Numpy do?
What will be the best option that covers all possibilities
Select one alternative:
Select one alternative:

- creates a two dimensional array with random numbers as values
- creates a two dimensional array with ones on the diagonal and zeros elsewhere. ✔
- creates a two dimensional array with zeroes as values.
- creates a two dimensional array with ones as values

Maks poeng: 1

9 Q9

Which of the following creates a Pandas Series with values [10, 20, 30] and index ['a', 'b', 'c']?

1. pd.Series([10,20,30], index=['a','b','c'])
2. pd.DataFrame([10,20,30], index=['a','b','c'])
3. pd.Series({'a':10,'b':20,'c':30})

Select one alternative:

- 1 & 3 ✔
- 1
- 1 & 2
- 3

Maks poeng: 1

10 Q10

Which of the following is an unsupervised learning task?
Select one alternative:

- Training a regression model
- Classifying images of cats and dogs
- Predicting stock prices
- Grouping customers by spending habits ✔

Maks poeng: 1

11 Q11

You are given the following DataFrames:

```
import pandas as pd
df_left = pd.DataFrame({
    "Product": ["A", "B", "C", "D"],
    "Price": [10, 20, 30, 40]
})
df_right = pd.DataFrame({
    "Product": ["B", "C", "E"],
    "Stock": [100, 150, 200]
})
```

Your goal is to
Create a DataFrame that contains:

- all products from df_left
- their prices
- matching stock information when available
- products without stock data should display NaN

Select one alternative:

- pd.merge(df_left, df_right, on="Product", how="outer")
- pd.merge(df_left, df_right, on="Product", how="left") ✔
- pd.merge(df_right, df_left, on="Product", how="right")
- pd.merge(df_left, df_right, on="Product", how="inner")

Maks poeng: 1

12 Q12

How will the plots be arranged?

```
Import matplotlib.pyplot as plt
fig = plt.figure()
fig.add_subplot(2,2,1).plot([1,2,3],[10,20,30])
fig.add_subplot(2,2,2).plot([1,2,3],[5,15,25])
fig.add_subplot(2,2,3).plot([1,2,3],[2,4,6])
fig.add_subplot(2,2,4).plot([1,2,3],[1,3,5])
plt.show()
```

Select one alternative:

- Four plots in a 2×2 grid, arranged like a square. ✔
- Four plots stacked vertically.
- Two plots side by side, the other two overlaid.
- One plot only with all lines combined.

Maks poeng: 1

13 Q13

What is the output of print statement?

```
df = pd.DataFrame({
    "a": [1, np.nan, 3],
    "b": [np.nan, 5, np.nan],
    "c": [7, 8, 9]
})
clean = df.dropna(subset=["a", "b"])
print(clean)
```

✓ 0.0s

Select one alternative:

	a	b	c
1	NaN	5	8

Empty DataFrame ✔

	a	b	c
0	1	NaN	7
1	NaN	5	8

	a	b	c
2	3	NaN	9

Maks poeng: 1

14 Q14

Which method converts a JSON string into a Pandas object such as a Series or DataFrame?
Select one alternative:

- pd.load_json()
- pd.to_json()
- pd.json_to_df()
- pd.read_json() ✔

Maks poeng: 1

15 Q15

You are merging a pandas DataFrame named customers (containing customer IDs and names) with a DataFrame named orders (containing order details and customer IDs).

Which type of merge operation ensures that every customer from the customers DataFrame is included in the result, even if they have no matching entries in the orders DataFrame?
Select one alternative:

- Inner Merge
- Right Merge
- Left Merge ✔
- Outer Merge

Maks poeng: 1

16 Q16

Given a 3D array 'arr' with shape (2, 3, 4), which command swaps the first and last axes?

1. arr.transpose(2, 1, 0)
2. arr.swapaxes(0, 1)
3. np.swapaxes(arr, 0, 2)

Select one alternative:

- 2
- 2 & 3
- 1
- 1 & 3 ✔

Maks poeng: 1

17 Q17

What is the main sign of overfitting for a machine learning model?
Select one alternative:

- Both training and test errors are low
- Both training and test errors are high
- Low training error but high test error. ✔
- High training error but low test error.

Maks poeng: 1

18 Q18

Out of the following data structures in python, which is **immutable**?
What will be the **best** option that covers all possibilities
Select one alternative:

Select one alternative:

- Tuple ✔
- Set
- List
- Dict

Maks poeng: 1

19 Q19

We have a DataFrame showing points scored by teams in different regions:

```
import pandas as pd
df = pd.DataFrame({
    'Region': ['North', 'North', 'South', 'South', 'South'],
    'Team': ['A', 'A', 'B', 'B', 'C'],
    'Points': [10, 15, 7, 5, 20]
})
```

We want the resultant dataframe out of df would look like this:

		Points
Region	Team	
North	A	25
South	B	12
	C	20

Which of the following commands produces this result?

Select one alternative:

- ○ df.groupby('Region')['Points'].sum()
- ○ df.groupby(['Region','Team']).sum() ✔
- ○ df.groupby(['Team','Region']).sum()
- ○ df.groupby(['Region','Team']).mean()

Maks poeng: 1

20 Q20

What is the purpose of using ‘header=None’ when reading a CSV file?

```
import pandas as pd
df = pd.read_csv("data.csv", header=None)
```

Select one alternative:

- ◯ It renames all columns to "None".
- ◯ It tells Pandas to skip the first row of the file.
- ◯ It tells Pandas that the file has no header row, so all rows are treated as data. ✔
- ◯ It creates an empty first row in the DataFrame.

Maks poeng: 1

21 Q21

You have a dataset with repeated entries in the "Name" column.

Which statement best describes the behavior of drop_duplicates(subset=["Name"], keep="first")?

Select one alternative:

- ◯ It only marks duplicate rows as True without removing them.
- ◯ It removes all rows where the "Name" appears more than once.
- ◯ It keeps the first occurrence of each unique "Name" and removes subsequent dupl ✔ es.
- ◯ It keeps the last occurrence of each unique "Name" and removes earlier duplicates.

Maks poeng: 1

22 Q22

Which function is commonly used as an activation function in hidden layers of neural networks?
Select one alternative:

- Mean squared error
- Linear
- Sigmoid, ReLU, or tanh ✔
- Softmax for regression tasks

Maks poeng: 1

23 Q23

What will be the output?

```
dates = pd.date_range(start="2025-01-01", periods=5, freq="D")
print(dates)
```

Select one alternative:

- DatetimeIndex(['2025-01-01', '2025-01-08', '2025-01-15', '2025-01-22', '2025-01-29'], dtype='datetime64[ns]', freq='D')
- DatetimeIndex(['01-01-2025', '02-01-2025', '03-01-2025'], dtype='datetime64[ns]')
- DatetimeIndex(['2025-01-01', '2025-01-02', '2025-01-03', '2025-01-04', '2025-01-05'], dtype='datetime64[ns]')
- DatetimeIndex(['2025-01-01', '2025-01-02', '2025-01-03', '2025-01-04', '2025-01-05'', dtype='datetime64[ns]', freq='D') ✔

Maks poeng: 1

24 Q24

What does this grouping do?

```
df = pd.DataFrame({
    "Numbers": [5, 12, 17, 24, 30]
})
grouped = df.groupby(lambda x: "Small" if x < 3 else "Large").sum()
print(grouped)
```

Select one alternative:

- Calculates the sum of the "Numbers" column only for rows where the index is less than 3, resulting in a single value of 34.
- Groups rows based on their index value, labeling indices 1 and 2 as "Small" and indices 3 and 4 as "Large", then sums the corresponding "Numbers".
- Groups the data by a custom column named "Small" or "Large" (depending on the numbers) and calculates the mean of each group.
- Groups rows based on the value in the "Numbers" column, labeling 5, 12, and 17 as "Small" and 24 and 30 as "Large", then sums the results.

Maks poeng: 1

25 Q26

What will be the output of this code?

```
x = 5
def outer():
    x = 10
    def inner():
        global x
        x = 15
    inner()
    print("Outer:", x)
outer()
print("Global:", x)
```

✓ 0.0s

Select one alternative:

- Outer: 10 and Global: 15 ✔
- Outer: 15 and Global: 10
- Outer: 15 and Global: 5
- Outer: 10 and Global: 5

Maks poeng: 1

26 Q26

Which Pandas method is used to save a DataFrame using Python's pickle format?
Select one alternative:

- df.write_pickle()
- df.save_pickle()
- df.to_pickle() ✔
- df.to_binary()

Maks poeng: 1

27 Q27

What is the result of a+b?

```
a = np.array([[1, 2, 3],[4, 5, 6]])
b = np.array([10, 20, 30])
```

Select one alternative:

- Raises a Value Error
- [11, 22, 33, 14, 25, 36]
- array([[11, 12, 13], [14, 15, 16]])
- array([[11, 22, 33], [14, 25, 36]]) ✔

Maks poeng: 1

28 Q28

What is the main purpose of L2 regularization (Ridge)?
Select one alternative:

- Removes outliers from the dataset.
- Increases training speed
- Normalizes the features.
- Penalizes large weights to reduce overfitting. ✔

Maks poeng: 1

29 Q29

Given the dataframe :
df = pd.DataFrame({'fruit': ['apple', 'banana', 'banana']})

How would you map apple → 1 and banana → 0?

Select one alternative:

- df['fruit'] = df['fruit'].replace({'apple': 1, 'banana': 0})
- df['fruit'] = df['fruit'].map({'apple': 1, 'banana': 0})
- df['fruit'] = df['fruit'].apply(lambda x: 1 if x=='apple' else 0)
- All above ✔

Maks poeng: 1

30 Q30

High bias in a model typically results in:
Select one alternative:

- Excellent performance on training set but poor on test set.
- Poor performance on training and test sets. ✔
- Excellent performance on test set but poor on train set
- Model variance

Maks poeng: 1

31 Q31

Which statement shifts the values backward by 2 periods?

dates = pd.date_range('2023-01-01 08:00', periods=4, freq='H')
s = pd.Series([5, 15, 25, 35], index=dates)

Select one alternative:

- s.shift(2, freq='H')
- s.shift(-2) ✔
- s.shift(2)
- s.shift(-2, freq='H')

Maks poeng: 1

32 Q32

What is the purpose of the rolling(window=3).mean() operation on a time series?
Select one alternative:

- It computes the average of every three consecutive values to smooth short-term fluctuations. ✔
- It shifts the time series forward by three periods before taking the mean.
- It calculates the cumulative mean of the first three values only.
- It randomly samples three values from the series and averages them.

Maks poeng: 1

33 Q33

What does backpropagation do in a neural network?
Select one alternative:

- Normalizes the input features.
- Converts categorical data into numeric.
- Computes the gradient of the loss function and updates weights. ✔
- Initializes the weights randomly.

Maks poeng: 1

34 Q34

Which of the following print statements will correctly print the variable loss rounded to 3 decimal places, if loss = 0.123456?

1. print(f"Loss value: {loss:.3f}")
2. print("Loss value: {:.3f}".format(loss))
3. print(f"Loss value: {loss}")

Select one alternative:

- 1& 3
- 2
- 1
- 1 & 2 ✔

Maks poeng: 1

35 Q35

Which type of plot is most suitable for showing the distribution of a single numeric variable?

Select one alternative:

- Stacked bar chart
- Scatter plot
- Bar chart
- Histogram ✔

Maks poeng: 1

36 Q36

You have this array:

```
arr= np.array([[1, 2, 3],
               [4, 5, 6],
               [7, 8, 9]])
```

Which command applied to the ’arr’ array, would produce this result?
result = array([6, 15, 24])

Select one alternative:

- arr.sum(axis=0)
- arr.mean(axis=1)
- arr.sum(axis=1) ✔
- arr.cumsum(axis=0)

Maks poeng: 1

37 Q37

What will be the output of the following print statement?

```
a = [1, 2]
b = [3, 4]
c = [5, 6, 7]

print (list(zip(a, b, c)))
```

Select one alternative:

- [(1, 3, 5), (2, 4, 6)] ✔
- [(1, 2), (3, 4), (5, 6)]
- [(1, 3, 5), (2, 4, 6), (3, 5, 7)]
- Error

Maks poeng: 1

38 Q38

What will be the result of the print statement?

```
my_dict = {'name': 'Alice', 'age': 25}
print(my_dict['city'])
```

Select one alternative:

- Raises a KeyError ✔
- Returns None
- Returns an empty string " "
- Returns {'name': 'Alice', 'age': 25, 'city': None}

Maks poeng: 1

39 Q39

You have the array:

```
import numpy as np
arr = np.array([[10, 20, 30],
                [40, 50, 60],
                [70, 80, 90]])
```

What will be the result of the indexing operation:
arr[[True, False, True]]

Select one alternative:

- array([[10, 40, 70], [30, 60, 90]])
- array([[10, 20, 30], [70, 80, 90]]) ✔
- array([[40, 50, 60]])
- None of the options

Maks poeng: 1

40 Q40

What will be the printed output of the following code?

```
try:
    value = float("abc") + 10
except TypeError:
    value = -1
except ValueError:
    value = 25
finally:
    value += 5

print(value)
```

Select one alternative:

- ○ 25
- ○ 30 ✔
- ○ -1
- ○ None

Maks poeng: 1